Pinak Limaye

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PROFESSIONAL SUMMARY

I am a software engineer who builds real-world software systems that combine AI capability with strong engineering fundamentals. I have delivered production solutions including VR-based medical platforms, secure licensing systems, and scalable cloud-hosted applications using Python, Unity/C#, Flask, and AWS.

I bring solid expertise in systems design, database architecture, and backend development, supported by advanced research experience in modern AI architectures such as diffusion models, transformers, and deep learning frameworks. I work effectively in Agile environments, collaborating with domain experts to translate complex problems into practical, reliable technical solutions.

I offer a rare combination of research depth, engineering versatility, and the ability to turn complex technical ideas into working, high-impact systems.

WORK EXPERIENCE

IT Contractor for 11 Monkeys — May 2026 — Present

Machine Learning Developer & Research Assistant

For this job I got contracted to develop a machine learning model for generating ancestors for paleontology fossils. To that end I have been working with multiple **3D generation models** such as **3D stable diffusion**, **SDFs**, **triplane prediction** and more. I have also worked on manuscripts for papers by editing information, validating references and scraping data from places like semantic scholar and google scholar. The job has me working in mainly **python**, writing scripts for both **Machine Learning** models and **Web Scraping**.

IT Contractor for High Fidelity Orthopaedics — February 2025 — April 2026

VR Development & Server Development

I got contracted to develop, design, and maintain a **virtual reality** solution for the orthopaedic company High Fidelity Orthopaedics. By working for HiFi, I learned and gained proficiency in the use of **Unity**, **C#**, **OpenXR**, and **MetaXR**. I also developed a **cloud server** using **Amazon** services such as **EC2**, **S3**, and **RDS**. The project allowed me to hone my skills for cloud development and server management. I implemented a **MySQL** database via **Flask**, background tasks via **Celery** and **Redis**, and hosted the server through **Nginx**. The server was primarily a means for storing and accessing data from our database and file server. I also set up the environment the server runs on manually inside the EC2. For the future I would use docker, however, I learned a great deal about environment management. I worked in an **Agile** type environment where we get feedback from orthopaedics and modify the solution based on the criticism, as a result the project is still ongoing.

IT Contractor — January 2021 — December 2024

Web Development & Visualisations

I got contracted to create visualisations of fossils and other datasets using a software called Vayu. During this time, I gained proficiency in **Flask**, **HTML**, **CSS** and **JavaScript** development. I maintained and developed a licensing system and license management website for Vayu. This work had me design a database for licensing purposes, along with creating a server for **user verification**. I also created some additional **anti-spoofing** measures to protect against potential piracy.

Telstra Health Internship — January — February 2020

Software Engineer Intern

In my internship at Telstra, I worked on code-cleanup and documentation for the software development team. I found errors by running **static code analysis** and developed a **Python** scraper to extract these errors or security concerns. I then used these results to analyse problems and fix common mistakes made by the team. I was also responsible for installation guides and documentation for the software the organization had used. From my internship, I learnt **Java** to an intermediate level and experienced working in a large team.

ASD Work Experience Program — September 2019

Cyber Security

I dabbled in multiple different cyber security divisions where I learned about **malware analysis**, hard drive forensics, **website pen testing**, electronics, **encryption and decryption** techniques and geographical analysis.

EDUCATION

Bachelor of Information Technology (Honours) — July 2023 — December 2024

University of Technology Sydney (GPA – 6.29, WAM – 81.57)

- Gained professional research experience by selecting and developing a novel solution in the field of **Timbre Transfer models**, focusing on audio processing and modern AI architectures such as transformers, diffusion networks, and VAEs,
- Authored a literature review and thesis on timbre transfer culminating in a **High Distinction** (64/70),
- Designed and implemented a **modification to a stable diffusion model**, enhancing its embedding module for timbre transfer, showcasing advanced skills in analysing and modifying large AI models at the code level,
- Studied **Advanced Mathematics** and **Signals Processing**, mastering concepts such as vector and matrix-based mathematics, statistics, discrete maths, and Fourier transformations for purposes of audio processing,
- Developed a 3D painting and sculpting application using three.js, applying technical and creative skills in computer graphics.

Bachelor of Information Systems — February 2020 — July 2023

University of Technology Sydney (GPA – 6.13, WAM – 81.88)

- Completed coursework in programming, **data analytics**, and **machine learning**, with a core focus on **Systems Design** and **Project Management** from requirements gathering to delivering a finished product,
- Proficient in utilizing reporting and design models such as **UMLs, ERDs, Use Case diagrams, Sequence diagrams**, and other system diagrams,
- Developed expertise in **requirement analysis**, **stakeholder analysis**, cost estimation, timeline creation, and **risk assessment** for projects,
- Led the **design phase of an automated checkout system** as part of a five-member team, including database design, software sourcing, and cost analysis,
- Gained proficiency in **Tableau** for reporting, complementing data analytics skills to derive insights using visual and analytical models,
- Studied additional subjects including **networking, cybersecurity, software development, and machine learning**:
  - Networking: Learned routing, the OSI model, proxy creation, error correction methods, and network protocols,
  - Cybersecurity: Acquired knowledge of intrusion methods, firewall setup, and advanced network security measures,
  - Database Design: Proficient in creating SQL databases up to 5NF, with practical experience in database normalization,
- Specialized in machine learning and data analytics with advanced proficiency in **Python programming** and libraries such as **PyTorch, TensorFlow, Pandas, Keras, PIL, and NumPy**,
- Familiar with the theory and implementation of models like **ANNs, CNNs, Diffusion models, YOLO models, and Q Learning models**, and experienced in using **KNIME** for analytical workflows,
- Worked on **Unix-based systems**, including **Linux** for web development and **Kali Linux** for Cybersecurity tasks.

Senior Secondary Education — February 2018 —February 2019

Gungahlin College (ATAR – 86.4)

- Majored in **Specialist Maths**, **Physics**, and **Double IT** (Web Development and Programming) at Gungahlin College.
- Studied **Markov chains** and the **Hidden Markov Model**, leading to the development of a **text-to-speech program** presented at PyCon.
- Designed and implemented a **basic interpreter** as part of a game teaching programming concepts, showcasing software development and education skills.
- Created a **LAN-based Dungeons & Dragons game** using web sockets, combining programming and web design expertise.
- Developed proficiency in **Python** programming, including **object-oriented** principles, as well as **JavaScript**, **Flask**-based web design, and **SQLite3** database design.

PROJECTS

Hidden Markov Based Text To Speech Model

I created a text to speech model using the Viterbi algorithm and a hidden markov model. I developed a novel method of text to speech using markov chains in a probabilistic model. I coded up the model in python and it heavily uses regular expressions and complex logic based thinking. In the model I create my own parser, use logic in order to utilize markov probabilities as confidence, and create a functional text to speech model based on character to character based probabilities.

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ASD Online Shopping System

This was a group project in which we had to create an online shopping system using Python, Flask and SQLAlchemy. We were a group of 5 and I worked on all the user related stuff. This included all the user management and user reviews along with all the SQL table creation and querying to go along with it. I designed the database for the whole system. I also created the login system, the verification system, account recovery, user editing, homepage, and the product review pages. We also set up Azure deployment for it, and deployed it to an Azure server along with test cases and an automated test system using pytest.

IoT Bay

This was a group project in which we had to create an online shopping system for purchasing electronic goods using Java. I primarily worked on user management, admin control and logging. For the project we used a view, model, controller design. I created all three for my designated parts. I also designed the database. The model was created using object oriented design that connected up to the database, and loaded in the entries as objects. We used JSP servlets in order to update web views, and database queries were done using get and post requests. All the validation was done with regular expression.

Pagerank TFIDF website retrieval

This was a group project in which we had to create a project using graph theory. We decided upon creating a page rank website retrieval feature. For this project I developed the pagerank algorithm, for which I used matrices and eigenvectors. I also developed the TF-IDF ranking of the pages. I used the two scores in order to determine page importance and reliability, along with relevance to the given search prompt. The novel aspect was the use of the Trie data structure in the TF-IDF calculation. The trie would allow the TF-IDF score to be calculated based on not only the relevant words in the search, but also give a relevance score to similar words. Also the trie data structure cut down the storage needed for page information to a fraction of the original space.

SKILLS

Programming Languages

Python • JavaScript • HTML/CSS • Java • SQL • C# • C++

Languages

English • Marathi • Hindi • Indonesian

Specialised skills

Data analytics • Machine Learning • Software development • Database Design • Systems Design • Visualisation • Project planning and design • Research • Cyber Security

AWARDS/HONOURABLE MENTIONS

- Thesis on **Audio Timbre Transfer** selected for submission into UTS Library **2024**,
- Selected for **UTS AI showcase 2023** – Created a YOLO model that detects distracted kids in classrooms,
- **Dean’s list for Information Technology 2022**,
- Second place in **YICTE ACT division 2019** – Given to my Viterbi Algorithm based text to speech model,
- **Pycon Presenter 2019** – Presented my Viterbi Algorithm based text to speech model,
- Got selected for and attended **National Computer Science School 2019**,
- **First Prize in Putt-Putt Coding Challenge** Senior Secondary/College Engineering Games **2017** – Created a mini golf game using Scratch.

REFERENCES

References available upon request.

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